Customer Subscription Prediction & Campaign Optimization

Predictive Analytics

By: **Lauryn Mwendwa**

Project Overview

This project transforms a traditional “call-everyone” marketing approach into a data-driven targeting strategy. By predicting customer subscription likelihood, the model enables businesses to focus on high-value leads, reduce wasted effort, and improve profitability. The outcome highlights how machine learning can drive smarter resource allocation and measurable business impact.

Objectives

Increase Conversion rates

Reduce call centre costs

Improve ROI of telephonic campaigns

Approach

- Built predictive model (Logistic + XGBoost)
- Used probability thresholding
- Focused on ROI optimization

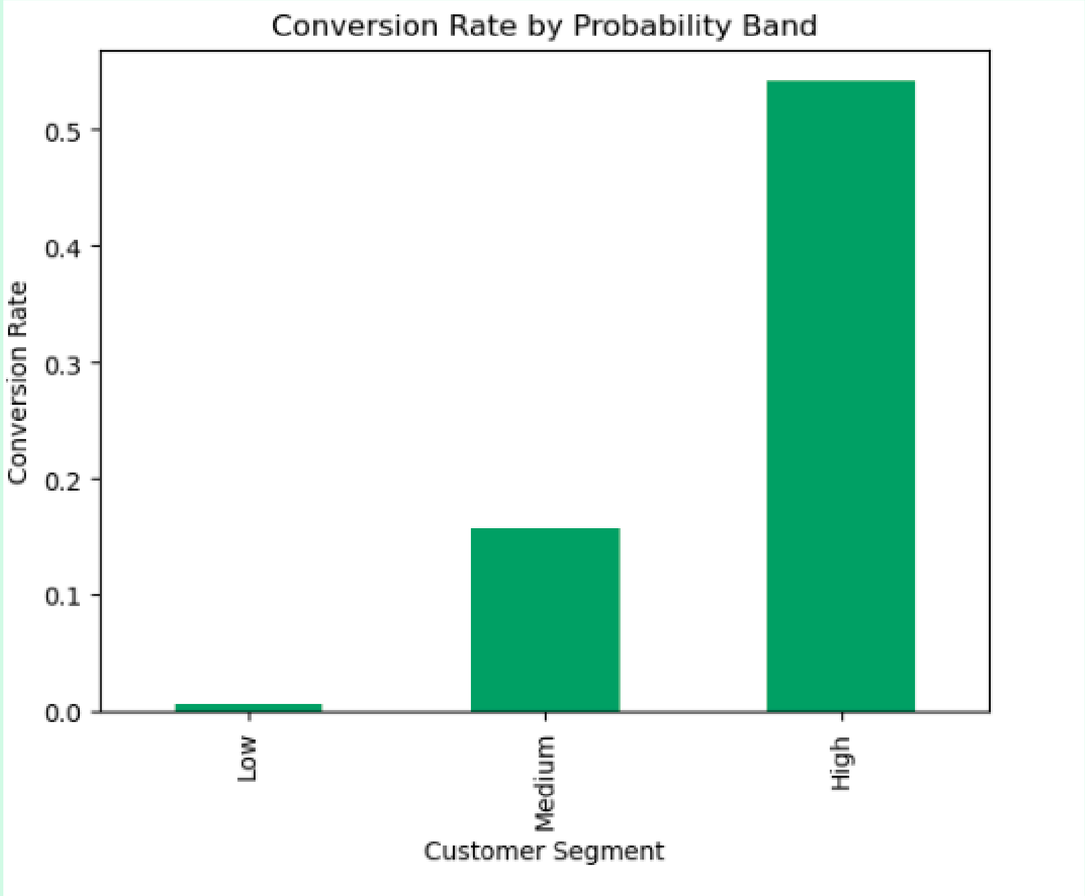
Model Performance

XGBoost Model

- Precision: 0.54
- Recall: 0.82

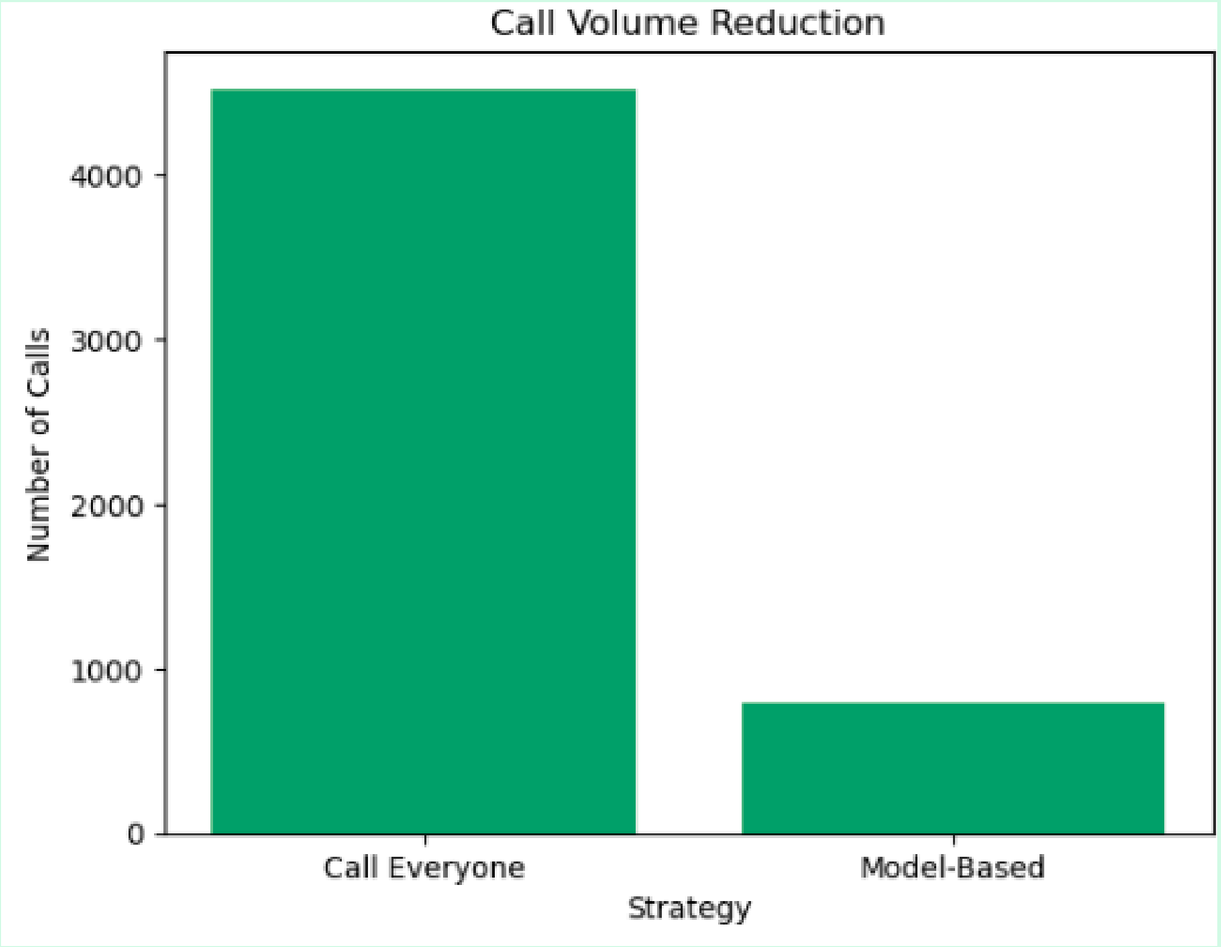
The model correctly identifies most potential subscribers while maintaining reasonable accuracy

Customers likely to Subscribe to Term Deposits



Customers in the high-probability group are far more likely to subscribe, making them ideal targets

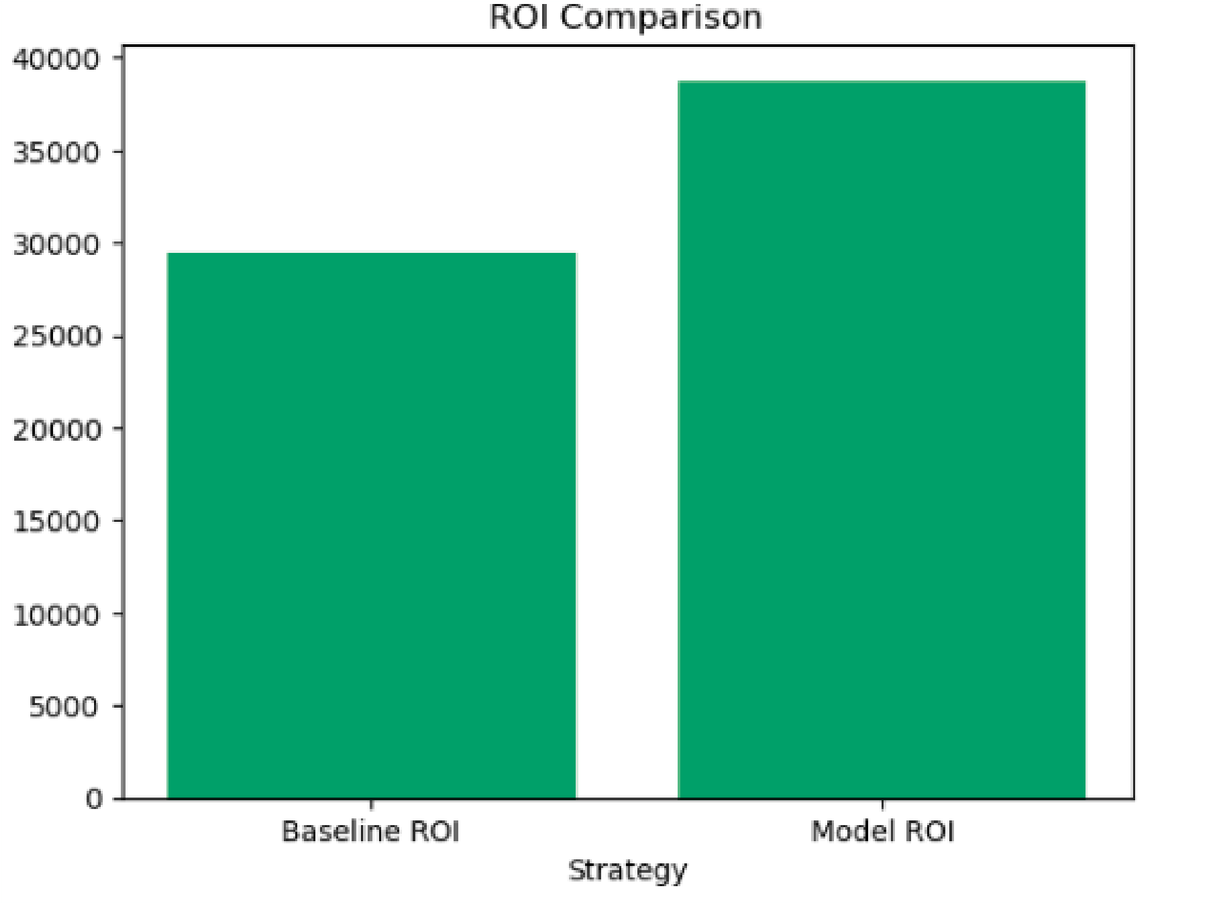
Operational Impact



- Calls reduced: 4521 → 791
- Saved: 3,730 calls

Using the model reduces outreach by over 80%(3730), significantly lowering operational effort.

Financial Impact



- Baseline ROI: 29,495
- Model ROI: 38,759
- Improvement: +9,264

Targeted campaigns generate higher profit despite fewer calls, proving efficiency over volume.

Recommendation

Call high-probability customers first.

Nurture medium group. Can send SMS/emails, then call if they engage

Ignore low group or very low-cost channels only

Insight

The model enables a shift from mass outreach to precision targeting, improving efficiency and profitability

Conclusion

Overall, the results show that targeted marketing enables businesses to move from broad, inefficient campaigns to focused, data-driven strategies. This not only reduces operational costs but also improves conversion rates and maximizes return on investment.